

Experiment No. 10

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Date:09/10/25

Roll No: 15

Grade:

Class: BE Comps (VII)

Sign:

AIM: Mini Project on MHT-CET Student Performance Analysis Dashboard 2025 using Power BI

GROUP MEMBERS:

1. Altamash Chougale – 05
2. Atharva Harane – 15
3. Darsh Kamble – 23
4. Sneha Singh – 64

THEORY:

The Maharashtra Common Entrance Test (MHT-CET) is one of the most important examinations for students aspiring to pursue engineering and pharmacy courses. Every year, lakhs of students appear for this exam, making it crucial to analyse and evaluate their performance to understand overall trends and insights. This project focuses on creating an interactive and comprehensive dashboard to visualize the performance of MHT-CET 2025 candidates.

The dashboard has been developed using Power BI, which enables dynamic and data-driven visualizations. It provides a clear representation of student performance based on multiple parameters such as gender, category, minority type, and percentile scores.

The dashboard includes several visual elements. A gender-wise distribution pie chart shows the participation ratio of male and female candidates. Minority type distribution helps to analyse the representation of candidates under different reservation categories. Overall, the MHT-CET Student Performance Dashboard serves as a powerful analytical tool that summarizes large volumes of exam data into meaningful insights. It not only helps in monitoring student outcomes but also assists stakeholders such as educators, institutions, and policymakers in making data-driven decisions. This project demonstrates the importance of data visualization in education and showcases how tools like Power BI can transform raw data into actionable knowledge.

TOOLS AND TECHNOLOGY USED:

1. Power BI → data visualization and dashboard design.
2. Excel / CSV → data source.

STEPS OF IMPLEMENTATION:

Step 1: Download and Install Power BI

1. Go to the official Microsoft Power BI website (<https://powerbi.microsoft.com>).
2. Download Power BI Desktop.
3. Install it on system by following the setup wizard.

Step 2: Prepare Dataset

1. Collect the MHT-CET dataset in Excel/CSV format.

Example fields:

- Gender
 - Category
 - Minority Type
 - EWS / TFWS
2. Clean the dataset (remove blanks, duplicates, check data types).

Step 3: Load Data into Power BI

1. Open Power BI Desktop.
2. Click on Get Data → Excel/CSV.
3. Select your dataset file and load it.
4. Verify the data in the Power Query Editor (apply transformations if needed).

Step 4: Data Processing & Cleaning

1. Rename columns for readability.
2. Data Cleaning
 - Null values in numeric fields (like HSC % or CET Percentile) were replaced with 0 to avoid calculation errors.
 - Empty text fields (like Category, Gender, or Minority Type) were also replaced with "Not Available" or 0 (depending on the data type).
 - Duplicate entries were checked and removed to maintain data accuracy.
 - Data types were corrected (percentile → decimal, category → text, application ID → number/text).

Step 5: Build Dashboard Visuals

1. Card Visual → Total Candidates, Total EWS Candidates, Total TFWS Candidates.
2. Pie Chart → Gender distribution, Minority distribution.
3. Line Chart → Category-wise candidate count.

Step 6: Formatting and Design

1. Add proper titles to each chart.
2. Apply consistent colours & themes (Power BI has built-in themes).
3. Insert slicers/filters (e.g., by Gender, Category).
4. Adjust layout → balance visuals (top-right-left-bottom properly arranged).

Step 7: Dashboard Title and Documentation

1. Add a project title: “*MHT-CET Student Performance Analysis – 2025*”.
2. Add text boxes for observations or key insights.
3. Save the file as .pbix.

Step 8: Export and Share

1. Export report as PDF (for submission) or publish it on Power BI Service (for online access).
2. Share with teachers, teammates, or evaluators.

INSIGHTS FROM MHT-CET PERFORMANCE DASHBOARD:

1. Gender Distribution

- The majority of candidates are male students, indicating higher participation compared to female students.
- Female participation is comparatively lower, which highlights the need for encouraging more female students to appear for competitive exams like MHT-CET.

2. Minority Type Distribution

- Most students belong to the General (Non-minority) category.
- A smaller proportion of candidates belong to minority categories such as Linguistic Minority (LM) and Religious Minority (RM).
- This shows that while reservation categories exist, their representation in the dataset is limited.

3. Category-wise Distribution

- Candidates from the Open category form the largest share, followed by OBC and other reserved categories.
- The presence of students across SC, ST, and NT categories highlights the diverse representation in MHT-CET.

4. Overall Performance Trend

- On average, students have performed well in CET, with a significant number scoring in the mid-to-high percentile range.
- The dashboard highlights both extremes — high achievers and weak performers — making it useful for performance tracking and improvement planning.

5. EWS and TFWS Representation

- The Economically Weaker Section (EWS) category reflects the government's initiative to provide opportunities for students from financially constrained backgrounds.
- A considerable number of candidates under EWS have qualified for admission through MHT-CET, indicating improved accessibility to higher education.
- The Tuition Fee Waiver Scheme (TFWS) category showcases the participation of meritorious students from low-income families who benefit from full or partial tuition fee exemptions.
- TFWS has enabled deserving candidates to pursue professional courses without financial burden, thus promoting inclusivity and academic merit.
- Together, EWS and TFWS categories highlight the social and financial inclusivity embedded within the MHT-CET admission framework.

CONCLUSION:

In this mini project, the MHT-CET Student Performance Analysis Dashboard 2025 effectively summarizes and visualizes student data to provide meaningful insights into overall exam trends. Using Power BI, the project highlights key patterns such as gender participation, category-wise representation, and performance variations across different groups. This analysis demonstrates the value of data visualization in education, supporting informed decision-making and promoting a more data-driven approach to student performance evaluation.

RESULTS:

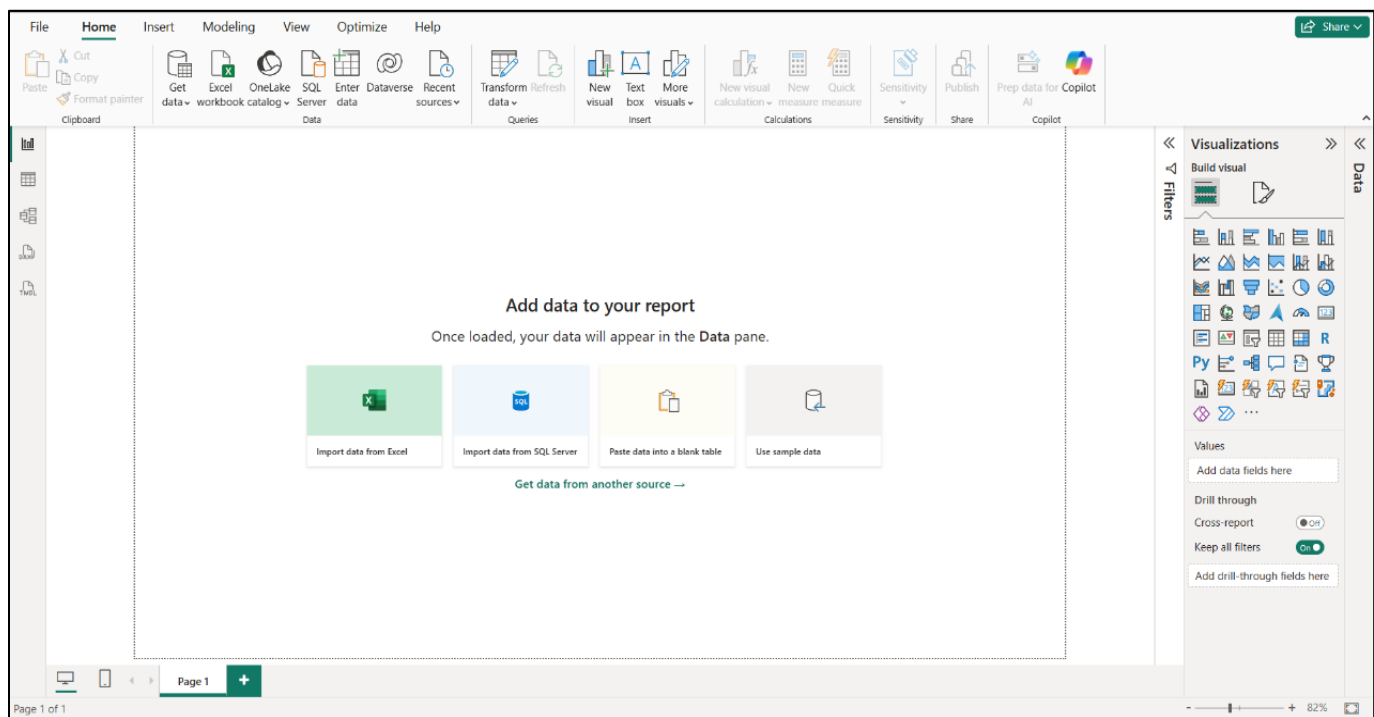


Table: ReplaceValue("#Replaced Value5",0,34,Replace.ReplaceValue,{"SSC(1f)Science(1f)X"})

Merit No	Application ID	Candidate's Full Name	Category	Gender	PWD / Def	EWS
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	EN25130336	PATTEWAR SHRISH NILESH	Open	Male	No	No
2	EN25262765	CHINMAY VIKAS CHAVAN	Open	Male	No	No
3	EN25261430	PATIL ANIL BALAJI	OBC	Male	No	No
4	EN25174886	ATHARV BHALCHANDRA SAHASRAB...	Open	Male	No	Yes
5	EN25181320	DUBEY AAYUSH ARVIND KUMAR	Open	Male	No	No
6	EN25171406	MUHAMMAD ALRAZA HANIF SHAIKH	Open	Male	No	No
7	EN25389743	TANAY CHETAN GADGIL	Open	Male	No	No
8	EN25292885	SARJE VAISHNAVI VITTHALRAO	SEBCS#	Female	No	No
9	EN25325566	ADITYA VASUDEO JUNGHARE	OBC	Male	No	No
10	EN25351284	SHREYA PANDEY	Open	Female	No	No
11	EN25155407	ARUSHI GOEL	Open	Female	No	No
12	EN25344507	ARNAV KUMAR	Open	Male	No	No
13	EN25363936	NIDHI ASHOK LOHANA	Open	Female	No	No
14	EN25176941	TANISH VISHAL JAIN	Open	Male	No	No
15	EN25368976	PATIL VISHWAJEET PRAMOD	Open	Male	No	No
16	EN25350923	AISHWARYA PRAKASH	Open	Female	No	No
17	EN25247794	KUNAL MAHENDRA PAREEK	Open	Male	No	No
18	EN25210462	LAIURKAR SUMIT SUNIL	OBC	Male	No	No
19	EN25184354	MANASI ARJUN FALAKE	SEBC	Female	No	No
20	EN25198011	GORSHETWAR ATHARV NAMDEV	Open	Male	No	Yes
21	EN25246607	BHAYANI CHANCHAL SANTOSH	Open	Female	No	No
22	EN25109862	SHAH RUSHIL PINAL	Open	Male	No	No
23	EN25205714	VEDANT AMOL VISPUTE	OBC	Male	No	No
24	EN25235077	GAHANE NIHAR BHOJPAL	OBC	Male	No	No
25	EN25383008	...	...	...	...	...

23 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 5:28 PM

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Merge Queries Append Queries Combine Files Combine

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Source_Page	Table_On_Page_Idx	Merit No	Application ID	Candidate's Full Name	Category	Gender
1	1	1	EN25130336	PATTEWAR SHRISH NILESH	Open	Male
2	1	1	EN25262765	CHINMAY VIKAS CHAVAN	Open	Male
3	1	1	EN25261430	PATIL ANIL BALAJI	OBC	Male
4	1	1	EN25174886	ATHARV BHALCHANDRA SAHASRAB...	Open	Male
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7	1	1	EN25389743	TANAY CHETAN GADGIL	Open	Male
8	1	1	EN25292885	SARJE VAISHNAVI VITTHALRAO	SEBCS#	Female
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21	1	1	EN25246607	BHAYANI CHANCHAL SANTOSH	Open	Female
22	1	1	EN25109862	SHAH RUSHIL PINAL	Open	Male
23	1	1	EN25205714	VEDANT AMOL VISPUTE	OBC	Male
24	1	1	EN25235077	GAHANE NIHAR BHOJPAL	OBC	Male
25	1	1	EN25297958	SOHAM AJAY SARATKAR	OBC	Male

25 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name Sheet1

APPLIED STEPS

Source

Navigation

Promoted Head...

Changed Type

Navigator

Display Options

BDA PROJECT DATA .xlsx [1]

Sheet1

Sheet1

Source_Page	Table_On_Page_Idx	Merit	Application	Candidate's Full Name
1	1	1	EN25130336	PATTEWAR SHI
1	1	1	EN25262765	CHINMAY VIKAS
1	1	1	EN25261430	PATIL ANIL BAL
1	1	1	EN25174886	ATHARV BHALC
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1	1	1	EN25176941	TANISH VISHAL
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1	1	1	EN25350923	AISHWARYA PF
1	1	1	EN25247794	KUNAL MAHEN
1	1	1	EN25210462	LAJURKAR SUM
1	1	1	EN25184354	MANASI ARJUN
1	1	1	EN25198011	GORSKETWAR

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Load

Transform Data

Cancel



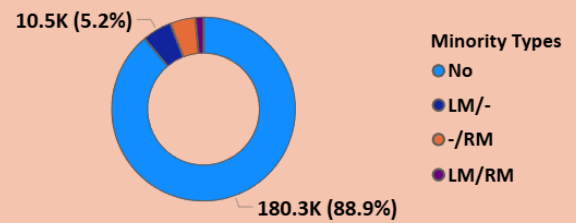
MHT-CET Student Performance Dashboard 2025

Total EWS
11.25K

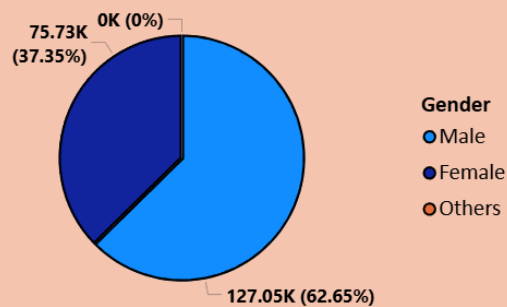
Total TFWS
63.00K

Total Candidate
202.78K

Count of Candidates by Minority Types



Count of Candidate By Gender



Candidates by Category

